

VNiVERSiDAD D SALAMANCA

Guide to conducting a systematic literature review (SLR)

Planning, searching, selection, synthesis, publication and responsible use of AI

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This guide explains, step by step, how to plan, conduct and publish a systematic literature review (SLR) or a systematic mapping study in engineering and computer science. It is intended for PhD students and researchers undertaking their first review.

How to use this guide

If this is your first SLR, read the chapters in order. If you already have a topic and need to get started, go straight to the phases (planning, search, selection, extraction and writing) and use the templates at the end. Each phase closes with the most common mistakes we see among PhD students. The two documents worth always keeping at hand are the process guidelines by Kitchenham and Charters [KC07] and the PRISMA 2020 reporting statement by Page *et al.* [PMB+21].

Quick roadmap

1. **Fundamentals**: what an SLR is, which types of review exist and which standards (PRISMA 2020, SEGRESS, Kitchenham, Petersen) you will have to comply with.
2. **Review phases**: planning and protocol, search, selection and quality, extraction and synthesis, writing and publication.
3. **Tools and AI**: which tool to use in each phase and how to use (and disclose) generative AI without compromising rigour.
4. **Practice**: a guided case and templates for the protocol, extraction form, screening sheet, checklist and schedule.
5. **References**: methodological guidelines, reporting standards and recommended readings in Spanish.

Contents

Part	Chapters
Fundamentals	*What is an SLR* · *Review types* · *Standards and guidelines*
Phases	*Planning* · *Search* · *Selection and quality* · *Extraction and synthesis* · *Writing and publication*
Tools and AI	*Tools* · *Generative AI in an SLR*
Practice	*Guided case study* · *Templates and checklists*
References	*References and resources*

Note

Tools and policies change. Check official sources and record date, version and applicable conditions. Methodological and visual revision: 16 October 2026.

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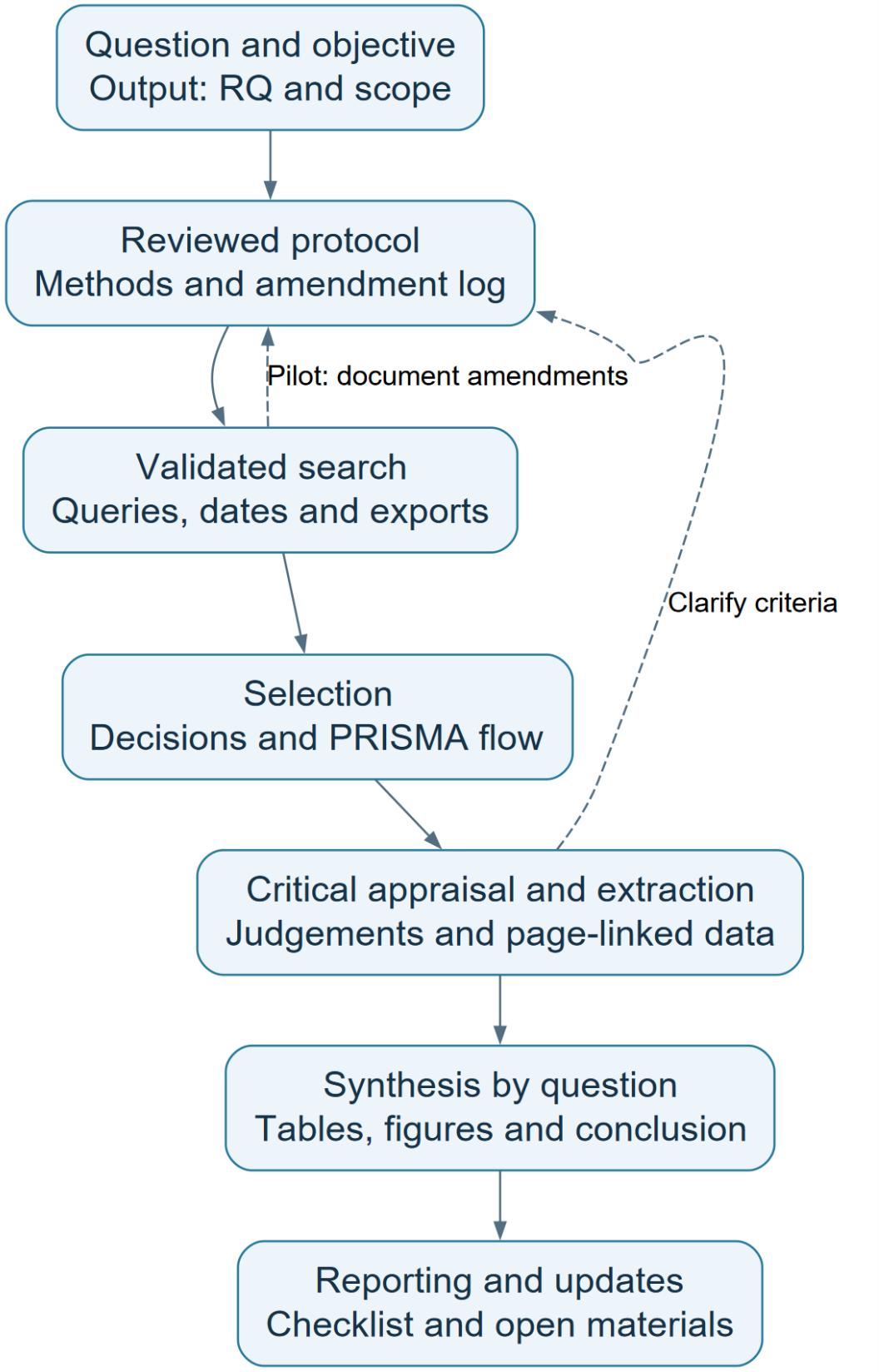


Fig. 1: Process and deliverables. Dashed arrows represent documented protocol amendments. Original figure, CC BY 4.0.

Part I

Fundamentals

CHAPTER 1

WHAT IS A SYSTEMATIC LITERATURE REVIEW

An SLR is a secondary study that identifies, evaluates and interprets all available evidence on a specific research question, following an explicit, unbiased and (to some extent) repeatable method (Kitchenham & Charters, 2007). Unlike original research, the unit of analysis is not people or systems but previous papers: the reviewer poses a question, collects data in the form of studies, analyses them and draws a conclusion.

Its goal is to state what is known about a topic, what has been investigated and what remains unresolved. Its decisive advantage over other reviews is that another researcher can replicate it and check whether they reach the same conclusion (Sáenz, 2001).

1.1 The four dimensions (Codina, 2015)

- **Systematic**: it is neither arbitrary nor subjective; it examines the best available scientific output using the best sources.
- **Comprehensive**: it uses information systems that give access to the bulk of high-quality international output and neither discards nor includes anything without explicit criteria.
- **Explicit**: it publishes the sources, the search criteria and the selection and exclusion criteria.
- **Reproducible**: being systematic and explicit, others can follow the steps and verify the results.

1.2 Origins and communities

SLRs originated in Medicine. The Cochrane Collaboration, founded in 1993, publishes reviews in the *Cochrane Database of Systematic Reviews*. The Campbell Collaboration is its sister organisation in the Social Sciences (Petticrew & Roberts, 2005).

In Software Engineering, the *Evidence-Based Software Engineering* (EBSE) community adapted the method from 2004 onwards, with the guidelines by Kitchenham (2004) and Kitchenham & Charters (2007), which remain the reference cited in most computer science SLRs. Since then the field has added its own guidelines for mapping studies (Petersen et al., 2008, 2015), *snowballing* (Wohlin, 2014), grey literature (Garousi et al., 2019) and reporting (SEGRESS, 2023). The standards section goes through them one by one.

1.3 Narrative review versus systematic review

Feature	Narrative review	Systematic review
Research question	Broad, unstructured	Structured, clear and specific
Literature search	Neither detailed nor systematic; high likelihood of bias	Detailed, systematic and explicit
Paper selection	No criteria; not reproducible	Explicit criteria applied uniformly to all papers
Quality appraisal	None	Critical appraisal of the methodological quality of each study
Synthesis	Subjective, qualitative summary	Based on study quality; often quantified
Interpretation	Sometimes evidence-based, often opinion-based	Generally evidence-based

Adapted from Martín Rodero (2014).

1.4 Systematic or systematized?

A systematic review can synthesise quantitative, qualitative or mixed-methods studies. **Meta-analysis is not required**: the synthesis method depends on the question, designs and available data. Narrative and thematic synthesis can also follow systematic methods.

Grant and Booth describe systematized reviews as work incorporating some elements of a systematic review, without necessarily meeting all of them. Explain the distinction through the methods actually used and their limitations, rather than the discipline or the absence of statistics [GB09, Cochrane].

1.5 Advantages and cost

A well-defined methodology reduces reviewer bias (although it does not protect against publication bias in the primary studies), shows whether a phenomenon is robust across different contexts and, with quantitative studies, allows data to be combined in a meta-analysis that detects effects small studies cannot see. The price is effort: an SLR demands much more time than a traditional review, and it is worth knowing this before you start.

CHAPTER 2

REVIEW TYPES AND WHICH ONE TO CHOOSE

Choose the design according to the objective: classify a field (mapping), clarify concepts and evidence types (scoping), or answer a focused question through critical synthesis (SLR). Meta-analysis is a synthesis technique that may form part of an SLR when studies are sufficiently compatible. The team's prior knowledge affects preparation, but does not determine the design.

Type	Typical question	What it produces	Quality appraisal	Reference guideline
Systematic review (SLR)	What is known about X? What solutions exist and how do they work?	In-depth synthesis of the evidence on a specific question	Yes, mandatory	Kitchenham & Charters (2007); PRISMA 2020; SEGRESS
Systematic mapping study (SMS)	How much has been published, by whom, where and on which subtopics?	Classification and count of the literature; maps, *bubble plots*, trends	Usually not, or light	Petersen et al. (2008, 2015); PRISMA-ScR
Meta-analysis	What is the combined effect of intervention X?	Aggregated statistical estimate (effect size)	Yes, with risk of bias	Cochrane Handbook; PRISMA 2020
Scoping review	What concepts, types of evidence and gaps exist in an emerging field?	Conceptual map of the area; often precedes an SLR	Optional	PRISMA-ScR (Tricco et al., 2018); JBI
Multivocal review (MLR)	What do academia **and** professional practice (blogs, reports, videos) say?	SLR that incorporates grey literature	Yes, with criteria for grey sources	Garousi et al. (2019)
Tertiary study / umbrella review	What do already published systematic reviews say?	Synthesis of reviews (the unit is the SLR)	Yes, with AMSTAR 2 or ROBIS	Kitchenham et al. (2009, 2010)
Rapid review	Same as an SLR, with fewer resources and a short deadline	Simplified synthesis; document the shortcuts	Simplified	Cochrane Rapid Reviews Methods Group

2.1 SLR versus systematic mapping study

The difference can be summarised as follows: a systematic review details the state of the art and seeks results that answer open questions; a mapping study gives a broad overview and identifies the volume of evidence, the types of research and the available results. In practice the term "systematic review" is used interchangeably for both, and both can be published in a journal and serve as a chapter of a thesis or master's dissertation.

Combining the designs can be useful when this matches the objectives and available resources: an initial mapping study tells you where the bulk of the literature lies, and a subsequent SLR delves into the subset that answers your question. Mapping questions are of the kind "how many studies were published per year?", "who are the most active authors?", "in

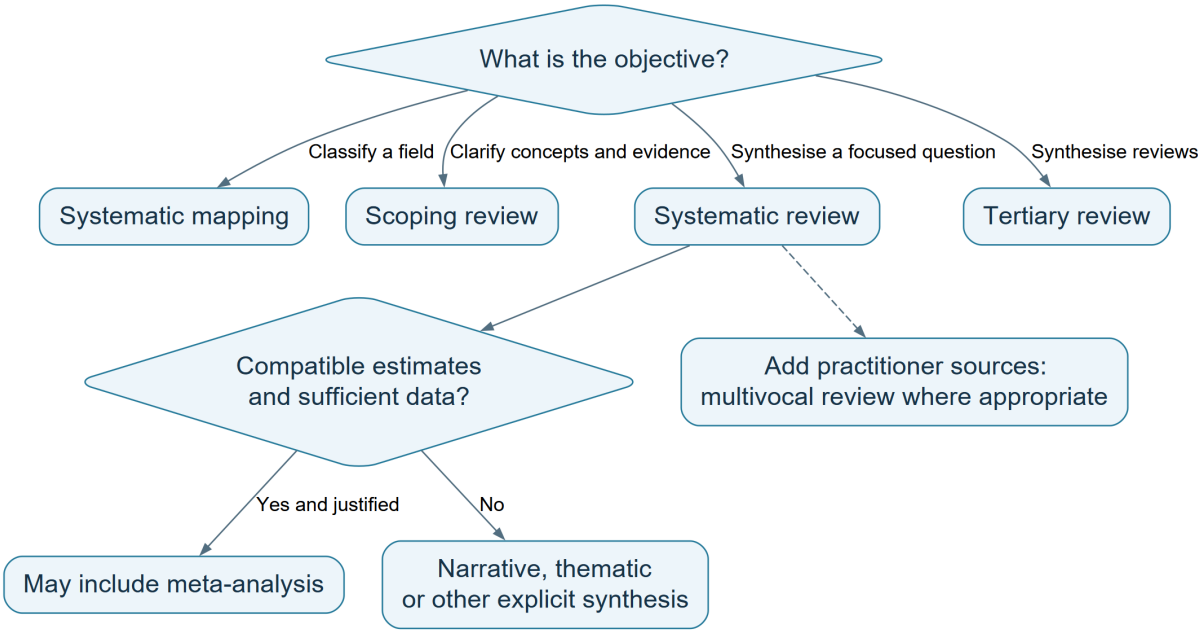


Fig. 2.1: Choose by objective. Designs can be combined with justification; meta-analysis is a synthesis technique. Original figure, CC BY 4.0.

which domains has it been applied?" or, in a review on LLMs in software engineering, "which models are used and for which tasks?" (Hou et al., 2024); SLR questions are of the kind "what solutions have been used for X?" and "how do they address X?".

2.2 Systematic review versus meta-analysis

An SLR answers a question by collecting and summarising all the empirical evidence with an explicit method; a meta-analysis is the use of statistical methods to combine the results of similar studies and obtain more precise estimates. Meta-analysis may form part of an SLR, but not every SLR contains one (Kitchenham & Charters, 2007; Mujica-Sequera, 2022). In engineering and computer science, meta-analysis is uncommon because primary studies rarely measure the same variables with comparable designs; when you do one, avoid *vote counting* (counting how many studies come out positive), a practice that the ACM SIGSOFT Empirical Standards explicitly discourage.

STANDARDS AND GUIDELINES YOU SHOULD KNOW

For an SLR in computer science today you need three things: a **process** guideline (Kitchenham & Charters for SLRs; Petersen for mapping studies), a **reporting** guideline (PRISMA 2020 or its SEGRESS adaptation) and, if you publish at an ACM/IEEE venue, the SIGSOFT **empirical standard**. The SALSA and PSALSAR frameworks help check that the design covers the four dimensions (systematic, comprehensive, explicit, reproducible).

Guideline	Year	What it is for	Where
Kitchenham & Charters, EBSE-2007-01	2007	Complete process of an SLR in software engineering: planning, conducting and reporting	Technical report
Kitchenham, Budgen & Brereton, *Evidence-Based Software Engineering and Systematic Reviews*	2015	Book that supersedes the 2004 and 2007 reports; covers primary studies, synthesis and knowledge translation	Routledge
Petersen et al., EASE	2008	Systematic mapping process: questions, search, screening, *keywording* of abstracts, extraction and map	PDF
Petersen, Vakkalanka & Kuzniarz, IST	2015	Update of the mapping guidelines after analysing how they were applied; concludes that the 2008 guidelines "are not sufficient on their own"	EBSE
Wohlin, EASE	2014	Backward and forward *snowballing* as a search strategy; EASE 2024 most influential paper award	EASE 2024 award
Garousi, Felderer & Mäntylä, IST	2019	Grey literature and multivocal reviews (MLR)	arXiv
PRISMA 2020 (Page et al., BMJ)	2021	27-item checklist for **reporting** a review and flow diagram; does not assess the quality of the review	prisma-statement.org
SEGRESS (Kitchenham, Madeyski & Budgen, IEEE TSE)	2023	Adaptation of PRISMA 2020 to software engineering secondary studies: SLRs, mapping studies, tertiary, qualitative	Checklist
ACM SIGSOFT Empirical Standards (Ralph et al.)	2020-2021	What an ICSE/FSE/EMSE reviewer will expect to see in a systematic review	Systematic Reviews standard

3.1 PRISMA 2020 in detail

PRISMA (*Preferred Reporting Items for Systematic Reviews and Meta-Analyses*) was created in 2009 and updated in 2021 (Page et al., BMJ 2021;372:n71). Its aim is for the review report to be complete and transparent; it can also be used to critically appraise published reviews, but it is **not** a methodological quality instrument. It consists of a 27-item checklist in seven sections, a 12-item checklist for the abstract and a flow diagram.

What changed with respect to 2009 and is worth bearing in mind:

- The search item requires the full strategies for **all** databases, registers and websites consulted, not just one example.
- You must state how many reviewers screened each record, whether they worked independently and which automation tools were used.
- The synthesis is broken down into six methods sub-items and four results sub-items.
- Items are added on certainty of evidence, conflicts of interest and availability of data, code and materials.
- The flow diagram has two arms: databases and registers, and "other methods" (citation searching, websites, organisations), which is added only if they were used. There are four official templates and a free web application (Haddaway et al., 2022) that generates the diagram in PDF, PNG or SVG.

There is an official Spanish translation of the statement (Rev Esp Cardiol 2021;74(9):790–799), and the PRISMA website also links the Spanish checklist and flow diagram.

3.2 PRISMA extensions useful in computer science

- **PRISMA-S** (Rethlefsen et al., 2021): 16 items for documenting the search (name of each database, full string, limits, filters, dates, peer review of the strategy, total records and deduplication). It is the best checklist for the search section of any SLR.
- **PRISMA-ScR** (Tricco et al., 2018): 20 essential items for scoping reviews; SEGRESS also recommends it for mapping studies.
- **PRISMA-P** (Moher et al., 2015): 17 items for writing the protocol.

3.3 SEGRESS: PRISMA adapted to software engineering

SEGRESS (IEEE TSE 49(3), 2023) takes the 27 items of PRISMA 2020, combines them with PRISMA-S, PRISMA-P and PRISMA-ScR, and indicates for each item which type of secondary study it applies to (quantitative SLR, mapping study, tertiary, qualitative, mixed). It replaces "quality assessment" with "risk of bias" and "threats to validity" with "limitations". If you are going to publish in IST, JSS or TSE, this is the checklist you should follow.

3.4 ACM SIGSOFT Empirical Standards

The standard for systematic reviews lists **essential** attributes (review type identified; step-by-step replicable search with the terms; clear selection criteria; detailed extraction and synthesis; coding scheme; chain of evidence; conclusions useful to practitioners) and **desirable** ones (protocol, searches and dataset as supplementary material; *snowballing* and preprints to mitigate bias; quality assessment defined a priori; two analysts with an agreement measure; PRISMA diagram; meta-analysis without *vote counting*). It names two antipatterns: the "shopping list" of papers without synthesis, and using the publication venue as a proxy for quality.

3.5 SALSA and PSALSAR

Grant & Booth (2009) propose the SALSA framework (*Search, AppraisaL, Synthesis, Analysis*) to check whether a review method meets the four dimensions. Mengist et al. (2020) extend it to PSALSAR by adding the **protocol** at the beginning and **reporting** (with PRISMA) at the end, in six steps: Protocol, Search, Appraisal, Synthesis, Analysis, Report. It is a good skeleton for the methodology section of your paper.

Part II

Review phases

CHAPTER 4

PHASE 1. PLANNING: FROM THE IDEA TO THE PROTOCOL

Planning ends with a written protocol that has been reviewed by someone other than you. Define questions, criteria, sources, search strategy, extraction and critical appraisal before the definitive search. Exploratory searches and pilots can refine the protocol. Later amendments may be necessary: record their date, reason, stage and impact, and distinguish prospective decisions from those informed by results.

4.1 Step 0. Is this review needed?

Ask yourself two questions (Kitchenham & Charters, 2007). First: is it necessary? Search Google Scholar, Scopus and Web of Science for "systematic review", "literature review", "mapping study", "survey" and "state of the art" together with your topic. If a recent SLR already exists, repeating it only makes sense if it is clearly biased or outdated because new studies have appeared (Petticrew & Roberts, 2005); otherwise, rethink the topic or turn your work into an update. Second: do you have the resources? An SLR takes months; you need time, access to the databases (through your university library), at least a second reviewer and a clear idea of who will use the results and how.

Before going any further, settle four things: scope, timeline, expected benefits and where you intend to publish, because the venue determines the length and the checklist you will have to comply with.

4.2 Step 1. Research questions

The question is the heart of the protocol: the title, the objectives, the search string and the extraction form all derive from it. It must be clear and specific, and answering it must require an explanation (never a yes or no). Check its suitability with the FINER acronym:

Criterion	Question to ask yourself
Feasible	Do I have the resources, access and time to answer it?
Interesting	Is it genuinely interesting to me and to my community?
Novel	Does it confirm, refute or extend previous findings?
Ethical	Could answering it harm anyone?
Relevant	Does it contribute anything to scientific knowledge or to practice?

Mapping questions categorise ("how many studies per year?", "which venues?", "which domains?", "what type of contribution: framework, tool, method?"). SLR questions go deeper ("what solutions have been proposed for X?", "how were they evaluated?", "what limitations do they report?"). It is advisable to number them (MQ1… for mapping, RQ1… for the SLR) and to make each column of the extraction form answer one of them.

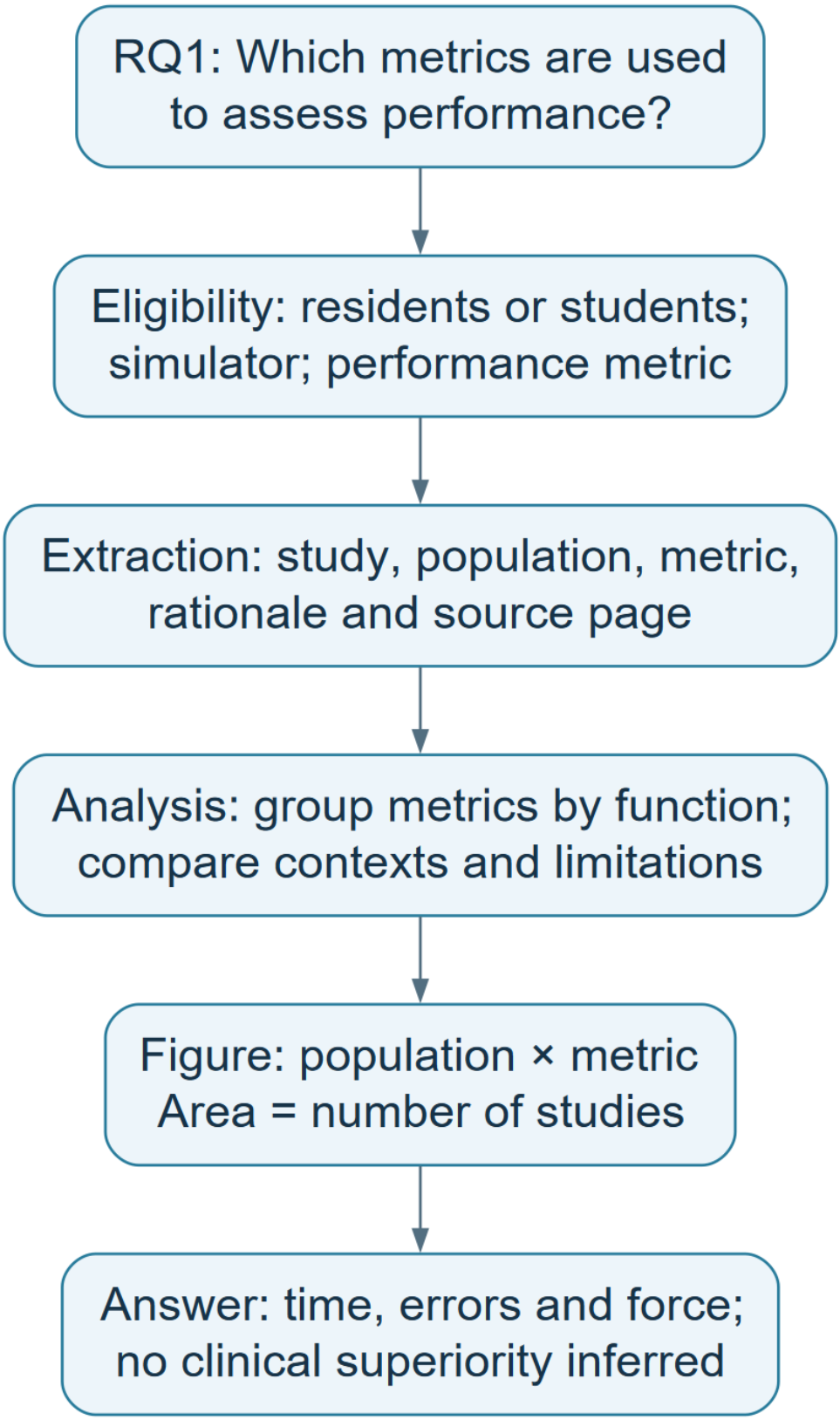


Fig. 4.1: From RQ1 to the conclusion in the fictional exercise. Each decision must be traceable to its source. Original figure, CC BY 4.0.

4.3 Step 2. Defining the scope with PICOC

PICOC (Petticrew & Roberts, 2005; Kitchenham & Charters, 2007) structures the question and, along the way, provides the search terms:

Element	Question	Example (Hou et al., 2024)
Populati	Who or what?	Primary studies that apply an LLM to a software engineering task (code generation, program repair, testing, vulnerability detection…)
Interven	What? How?	Large language models (GPT, Codex, CodeBERT, LLaMA…) and the techniques used to adapt them: *prompting*, fine-tuning, context retrieval
Compar	Compared with what?	Across model architectures, data strategies and optimisation techniques; there is no control group
Outcom	What is to be achieved or improved?	Which models are used, how data are prepared, how performance is optimised and evaluated, and in which tasks they work best
Context	In which organisation and circumstances?	Academic research in software engineering, including arXiv preprints, from 2017 to January 2024

In computer science the "population" is usually a type of system, technique or role (developers, programming students), and the comparison often does not exist; make this explicit. For qualitative reviews you can use SPIDER (*Sample, Phenomenon of Interest, Design, Evaluation, Research type*).

4.4 Step 3. Writing the protocol

A minimal protocol has eight elements: questions and objectives, inclusion criteria, exclusion criteria, databases and search engines, search terms, search and extraction procedure (iterated in stages), quality assessment and synthesis of results. Also add: the number of reviewers and how you will resolve disagreements, the extraction form, the synthesis plan, the timeline and the dissemination strategy. PRISMA-P (17 items) serves as a checklist; the template at the end of this guide follows it.

Ask your supervisor or a colleague to review the protocol before you carry it out. Kitchenham & Charters list this as an activity of its own within planning ("evaluating the protocol"), and it is the cheapest moment to fix a poorly framed question.

4.5 Step 4. Registering or publishing the protocol

Registering the protocol before execution improves transparency and allows comparison of planned and completed methods; it does not guarantee absence of bias. PRISMA 2020 asks authors to report registration and protocol access, or state that neither exists. Public deposit and prospective registration are not equivalent. Options:

- PROSPERO (University of York, free): only accepts reviews with outcomes relevant to human health; it is not suitable for most SLRs in computer science or education, and it excludes scoping reviews. Relaunched in March 2025 with automatic publication after the authors' approval.
- OSF Registries (free, with DOI, any discipline): the most widely used option outside health; it offers a "Generalized Systematic Review" template suitable for SLRs, scoping reviews and meta-analyses.
- INPLASY (check current fees and conditions) and Research Registry (paid): accept any field.
- Open deposit in Zenodo, Figshare or arXiv: common practice in software engineering, with a persistent identifier and date, although without review; Zenodo and Figshare support DOIs, whereas arXiv supplies an arXiv identifier.

Only 31.5% of the 537 secondary studies in software engineering published between 2013 and 2023 share their artefacts (protocol, searches, data), and barely 12% do so in a permanent repository (Huotala, Kuutila & Mäntylä, 2025). Doing it properly sets you apart.

4.6 Team and timeline

An SLR is not a one-person job: PRISMA 2020 asks you to state how many reviewers screened and whether they did so independently, and the Empirical Standards consider it desirable to have two analysts with a measure of agreement. For a PhD student, the usual arrangement is for the supervisor or a colleague to act as second reviewer, at least on a sample. As for timing, allow 4 to 9 months for a complete SLR in an area with several hundred candidates; the most underestimated phase is full-text reading and extraction.

4.7 Common mistakes in this phase

Common mistakes in this phase

- Starting to search without a protocol and "reconstructing" it afterwards.
- Questions so broad that the answer fits in a narrative review ("what is known about AI in education?").
- Not checking whether an SLR on the topic already exists.
- Changing the inclusion criteria midway without documenting or justifying it.
- Not deciding from the outset in which venue the review will be published and which checklist it requires.

CHAPTER 5

PHASE 2A. SEARCH: SOURCES, STRINGS AND DOCUMENTATION

The search must be replicable: another researcher, with your string, your database and your date, should obtain (almost) the same records. Only 43% of 311 analysed software engineering SLRs publish reusable strings, and fewer than 5% reproduce their results (Li, 2021); do not become part of that statistic.

5.1 Where to search

Choose sources for topical coverage, study types and access, and justify the combination. Consult a search specialist where possible. There is no universal sufficient number of databases. Indexes and platforms overlap; retain provenance before deduplication [R+21b].

Source	Coverage and use
Scopus	Multidisciplinary index; journals and conferences
Web of Science	Report the collections searched
IEEE Xplore	Engineering and computing
ACM Digital Library	Distinguish full-text collection and bibliographic index
SpringerLink / ScienceDirect	Publisher platforms; justify additional coverage
ERIC	Education; free text and thesaurus
Google Scholar	Complement; report limits and ranking
dblp	Computing metadata; no abstracts
OpenAlex / Semantic Scholar	Check coverage, fields and API conditions

Export limits, fields and syntax change. Check official documentation when searching and log date, format, batches and exported total. Full-text retrieval differs from identifying records.

5.2 Building the search string

1. **Derive the concepts from PICOC**: usually population, intervention and outcome; context rarely goes into the string.
2. **Expand each concept with synonyms**: derived terms, spelling variants (*modelling*/*modeling*), acronyms (HCI), terms in other languages where appropriate, and descriptors from the database thesaurus (MeSH in MEDLINE, ERIC thesaurus, IEEE Thesaurus, ACM CCS).
3. **Join synonyms with OR and concepts with AND**; avoid NOT except to exclude a clearly unrelated domain, because it removes relevant records that mention the term in passing.

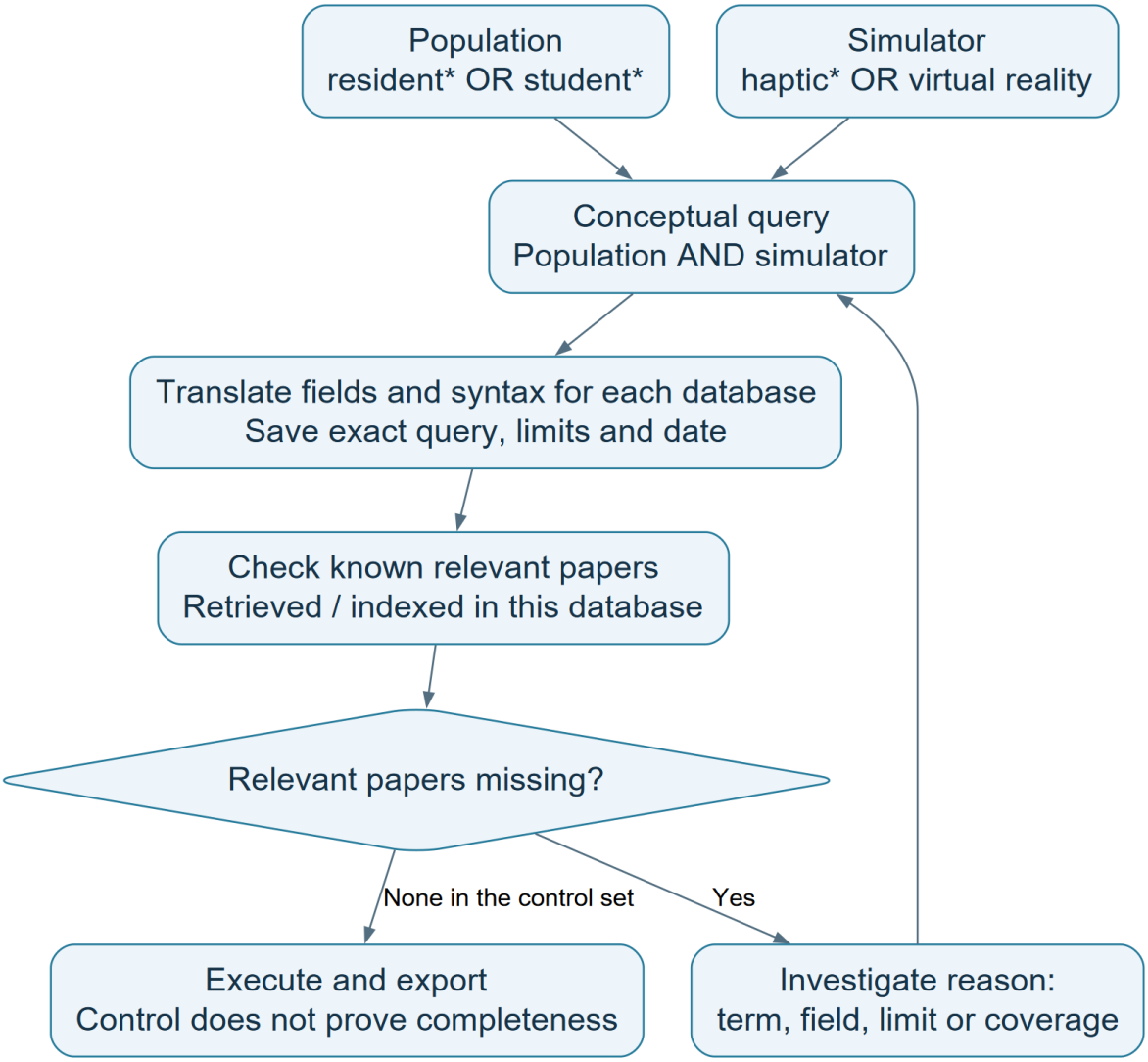


Fig. 5.1: Building and checking a search. Conceptual expressions: adapt syntax, fields and truncation to each platform. Original figure, CC BY 4.0.

4. **Use parentheses** to set precedence and quotation marks for exact phrases; truncation (*) broadens the search (*analy** covers *analysis*, *analyse*, *analytics*).
5. **Test the string against a control set**: 5 to 10 papers that you know must appear. If any is missing, revise the terms.
6. **Adapt the syntax to each database** and save the exact version.

Example from Hou et al. (2024), a systematic review on the use of large language models (LLMs) in software engineering tasks, published in *ACM Transactions on Software Engineering and Methodology* with all its materials openly available. The string combines two concepts, each with a long list of synonyms joined by OR:

```
Concept 1 (software engineering tasks):
  "Software Engineering" OR "Software Development" OR Program* OR "Code Generation"
  OR "Code Completion" OR "Code Summarization" OR "Bug Detection" OR "Program Repair"
  OR "Test Generation" OR "Vulnerability Detection" OR ...

Concept 2 (language models):
  LLM OR "Large Language Model*" OR "Language Model*" OR LM OR PLM OR Pretrained
  OR "Pre-trained" OR GPT OR ChatGPT OR Codex OR BERT OR CodeBERT OR T5 OR LLaMA OR ..
 ↪.

Final string:  (Concept 1) AND (Concept 2)
```

The search was run on seven sources (IEEE Xplore, ACM Digital Library, ScienceDirect, Web of Science, SpringerLink, arXiv and dblp), limited to January 2017–January 2024, and returned more than 218,000 records, which the screening phases reduced to 395 included studies. Three practices from that review are worth copying: before the automated search, they performed a manual search in six reference venues (ICSE, FSE, ASE, ISSTA, TOSEM and TSE) and used it to build a control set of 51 papers (*quasi-gold standard*) to validate the string; they included arXiv and dblp as sources complementary to the traditional libraries; and they published the strings adapted to each database, the lists of papers and the screening decisions in a public repository, so that anyone can replicate the search.

The strings for each database must be identical or equivalent; otherwise, the results are not comparable. The example by Ferreras-Fernández et al. (2016) shows an additional good practice: recording in a table each numbered query (#1, #2…), the field (title, topic), the filters (years, languages) and the number of results, and combining them at the end (#1 OR #2 OR #3 OR #4).

5.3 Snowballing

The guidelines by Wohlin (2014) propose starting from a heterogeneous initial set (different authors, years and venues) and reviewing the reference lists (*backward*) and the citations received (*forward*, with Google Scholar or Scopus) until an iteration yields no new papers. Wohlin et al. (2022) show that the hybrid Scopus + *snowballing* strategy finds around 30% more relevant papers than database searching alone. In the PRISMA diagram these papers go in the “other methods” arm.

5.4 Grey literature

In topics where professional practice is ahead of academia (DevOps, LLMs in programming, commercial educational tools), a multivocal review adds technical blogs, reports and documentation. Garousi et al. (2019) provide criteria for deciding whether to include it and how to assess its quality (authorship, date, methodology, citations). Document Google searches with the date, the number of pages reviewed and the stopping criterion.

5.5 What to document (PRISMA-S)

For each source: name and platform (Scopus via Elsevier), the full string exactly as executed, fields, filters and limits (years, language, document type), date of execution, number of records returned and deduplication method with the number of duplicates removed. If you update the search before submitting the paper, document both runs. Store everything in the supplementary material (Zenodo or OSF).

5.6 Common mistakes

Common mistakes

- A different string for each database without a table of equivalences.
- Reporting "Google Scholar was searched" without a date, without the number of results reviewed or a stopping criterion.
- Not keeping the original export (RIS/BibTeX) from each database.
- Forgetting conference proceedings in fields where they are the main venue (software engineering, HCI).
- Limiting by date or language without justifying it in the protocol.

5.7 Validation and execution log

Not every PICOC component needs an AND block. Adding outcomes or metrics poorly described in abstracts can lose relevant studies. Compare variants, inspect what each block adds or removes and retain the history. The figure uses two conceptual blocks; it is not a ready-to-run query for every platform.

For each database, the control denominator is the number of known papers indexed there and meeting its limits. Retrieving all of them does not establish completeness. Log why each missing paper was missed and avoid tuning only to their wording. Publish the full query without "…": the shortened Hou query is illustrative, not reproducible.

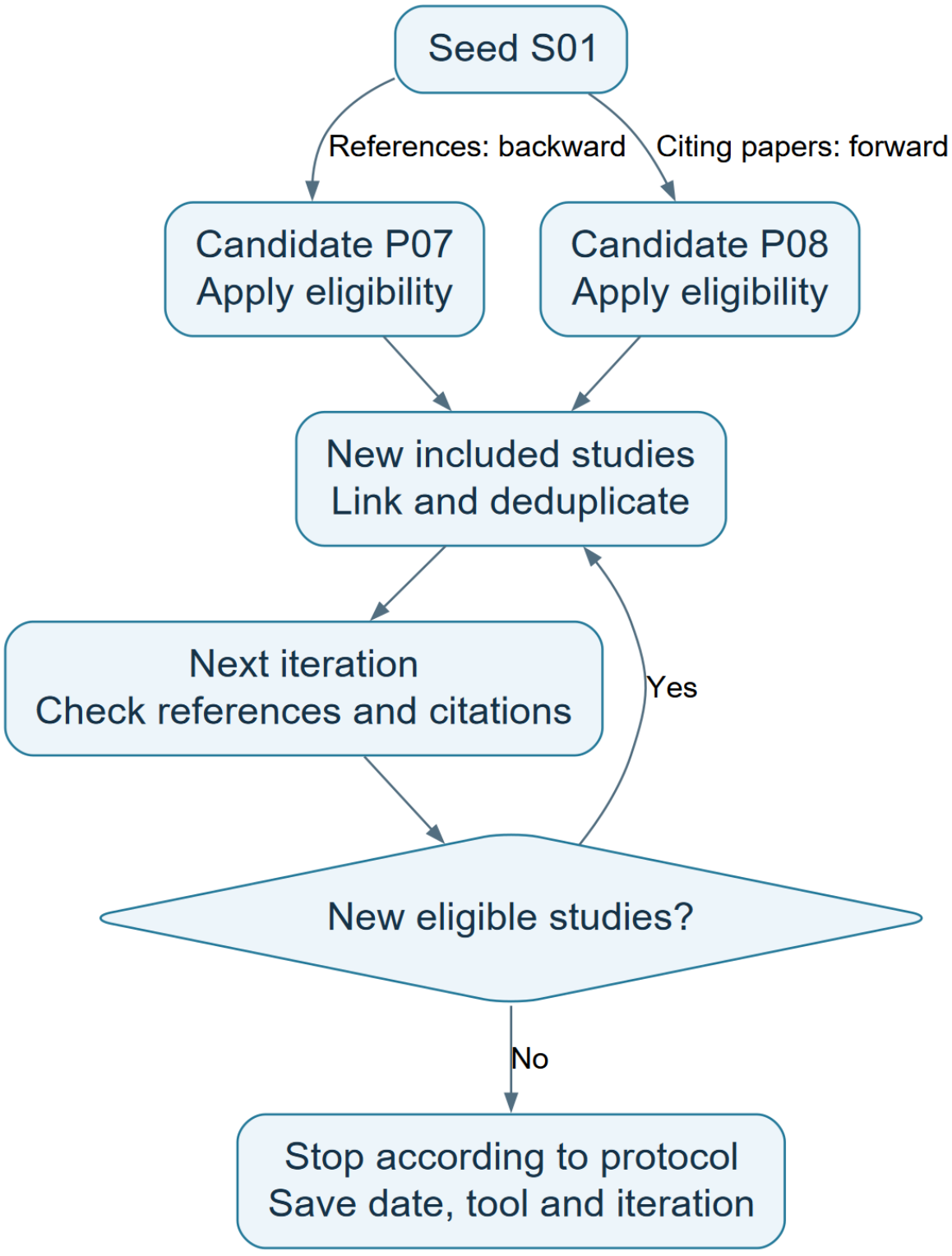


Fig. 5.2: Citation searching and iterations. Arrows are discovery steps, not citation direction. Conceptual example independent of the dataset. Original figure, CC BY 4.0.

PHASE 2B. STUDY SELECTION AND QUALITY ASSESSMENT

Selection turns hundreds or thousands of records into the dozens of primary studies that will answer your questions. It is done in phases, with criteria fixed in the protocol and, whenever possible, with more than one reviewer.

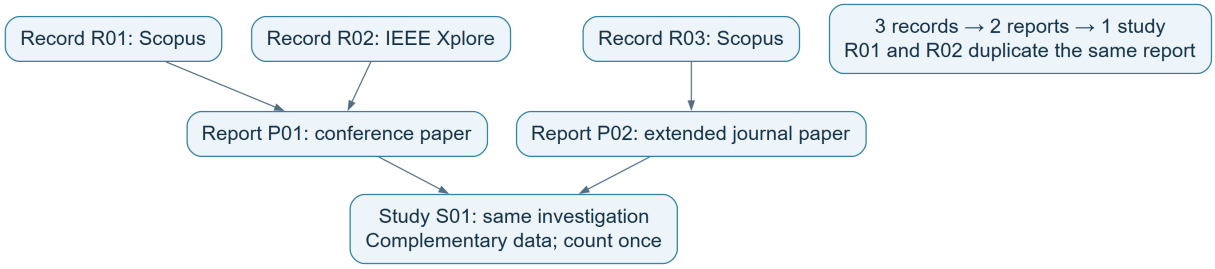


Fig. 6.1: Records, reports and studies: conceptual example independent of the dataset, based on PRISMA 2020 definitions. Deduplication and report linking are different operations. Original figure, CC BY 4.0.

6.1 Inclusion and exclusion criteria

They are defined from the research questions, plus some generic criteria (language, date, document type, peer review). They must be clear and unambiguous, and any later change must be justified and documented; avoid result-driven changes; when an amendment is necessary, document its rationale, timing and impact. Inclusion criteria are chained with AND and exclusion criteria with OR: a single exclusion criterion is enough to discard a paper.

Example (Hou et al., 2024, review on LLMs in software engineering): IC1 the paper states that it uses an LLM **and** IC2 the study addresses a software engineering task **and** IC3 the full text is accessible. Exclusion criteria: EC1 short papers (fewer than 8 pages) **or** EC2 duplicates or different versions of the same study by the same authors **or** EC3 books, theses, monographs, *keynotes*, panels or venues without full peer review **or** EC4 tool demonstrations and editorials **or** EC5 workshop or doctoral symposium papers **or** EC6 grey literature (technical reports) **or** EC7 not written in English **or** EC8 mentions LLMs without describing the technique used **or** EC9 uses software engineering methods to improve LLMs rather than LLMs for software engineering tasks. Note that the last three are not generic but topic-specific: they are the ones that truly separate the relevant study from the one that merely touches on the topic. With these criteria, more than 218,000 records were reduced to 810 after title and abstract screening, to 594 after full text and to 382 after quality assessment; *snowballing* added 13, for a total of 395.

Common generic criteria: date (justified: the emergence of a technology, a previous review), language, publication type (editorials, extended abstracts, posters and non-reviewed chapters are usually excluded), population or context (educational level, type of organisation), study design (empirical only, with evaluation only) and access to the full text. Link reports of the same study: they may provide complementary data. Specify the source for each outcome and avoid counting participants or effects twice. Record unavailable reports separately from eligibility exclusions in PRISMA.

6.2 The screening phases

1. **Run the queries** and export the records from each database (RIS or BibTeX), noting the date.
2. **Remove duplicates** with the reference manager or a dedicated tool; record the number removed.
3. **Screen by title and abstract** applying the criteria. When in doubt, pass the paper on to the next phase; here it is better to be inclusive.
4. **Screen the full text**, recording the reason for excluding each discarded paper (PRISMA requires listing them with their reasons).
5. **Add the papers from *snowballing*** and repeat the screening on them until no new ones appear.

The number of phases depends on the volume of candidates and the number of reviewers. Exclusion rates vary widely: estimate workload through a pilot of your own corpus. Before screening in earnest, calibrate: the reviewers independently screen a sample of 50 to 100 records, compare their decisions and refine their interpretation of the criteria.

6.3 Inter-reviewer agreement

Independent screening by two people reduces errors; disagreements are resolved by consensus or a third person. If only part can be duplicated, justify sample size and selection, cover difficult decisions and report the limitation. **Twenty per cent is a planning example, not a validated minimum or a guarantee against bias.** Expand or repeat calibration when criteria change or important disagreements arise.

Report how many records both people assessed, their decision table before consensus, raw agreement and, where appropriate, kappa with its uncertainty. Conventional kappa labels are only indicative and depend on prevalence and margins. A sample dominated by exclusions may show high agreement while hiding mistakes in relevant studies.

Test-retest assesses consistency within one reviewer, but cannot replace independent assessment. An LLM is an automated tool and must be described and validated as such, even when it supplies a second proposed decision.

6.4 Critical appraisal: quality, risk of bias and certainty

These related concepts answer different questions:

- **Methodological quality:** appropriateness and transparency of a study's design, conduct and analysis.
- **Risk of bias:** the possibility that its methods systematically distort a result.
- **Certainty of evidence:** confidence in a body-of-evidence conclusion for a specific outcome. GRADE considers risk of bias, inconsistency, imprecision, indirectness and publication bias [SBA+23].

Choose the tool for the included designs and follow its manual. Not every tool permits summing scores: **MMAT discourages an overall score**, and AMSTAR 2 uses critical weaknesses rather than a simple total [HPF+18, SRW+17]. Retain each domain judgement, its rationale and the supporting page or passage.

Instrument	For which type of study	Notes
Dybå & Dingsøyr (2008)	Empirical software engineering studies	11 yes/no criteria: is it research rather than *lessons learned*?, aims, context, design, recruitment, control group, data collection, analysis, reflexivity, findings, value. The most widely used in software engineering
Kitchenham & Charters (2007), appendix	Any empirical SE study	Questions by design type (experiment, case study, survey)
CASP	Qualitative studies, trials, cohorts, reviews	Free; originated in health, adaptable
MMAT 2018	Qualitative, quantitative and mixed-methods studies	Five categories, five criteria each; useful in education
AMSTAR 2	For appraising systematic reviews (tertiary studies)	16 items; overall rating of high, moderate, low or critically low
ROBIS	Risk of bias in systematic reviews	Four domains: eligibility, identification and selection, data collection, synthesis

RoB 2 targets results from randomised trials; GRADE addresses a body of evidence by outcome. Neither should automatically be applied to every computing study.

The criteria used by Hou et al. (2024) illustrate decisions in a published review, rather than a universally recommended instrument. In particular, venue prestige cannot replace methodological appraisal. Prespecify and justify any adaptation or appraisal-based exclusion. Do not turn a quality total into statistical weights: meta-analysis weights depend on the model and the precision of estimates.

Explain how appraisal informs synthesis: identify limitations, compare conclusions after removing studies with major concerns, or moderate confidence in findings. Appraisal may be omitted with justification in descriptive mapping; conclusions about effectiveness require critical assessment.

Two people should compare judgements and resolve disagreements. Test-retest assesses one person's consistency and is not equivalent to independent appraisal.

6.5 Common mistakes

Common mistakes

- Being too strict when screening by title and abstract, and losing studies that the abstract describes poorly.
- Not recording the reason for exclusion at the full-text stage.
- Not measuring inter-reviewer agreement, or measuring it and not reporting it.
- Using the publication venue or the number of citations as a substitute for quality.
- Inventing your own quality checklist when a validated one exists.

For an actual review, use the official PRISMA 2020 template for the review type and sources used. This is a teaching adaptation, not an unchanged official template. Count records during identification/screening, reports during re-

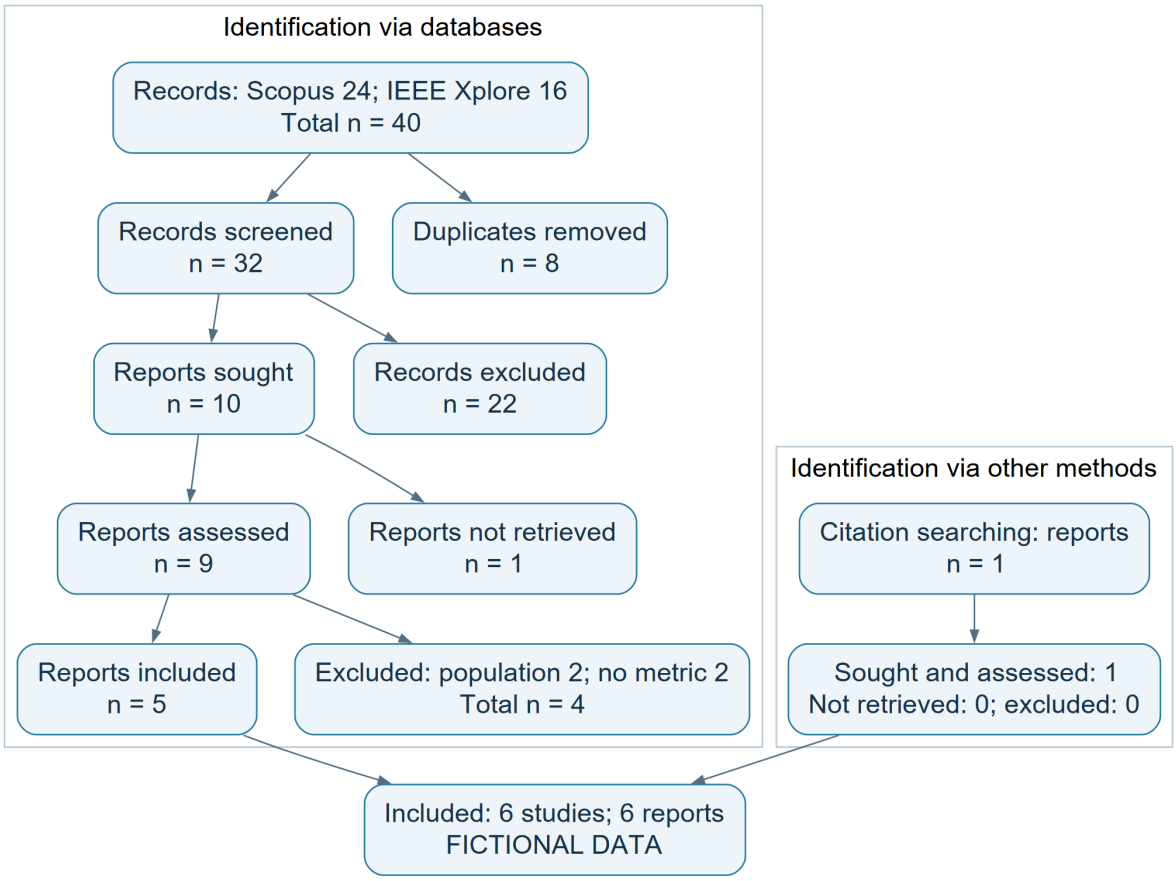


Fig. 6.2: Teaching flow adapted from PRISMA 2020 (Page et al., 2021), CC BY 4.0. Fictional data: 40 − 8 = 32; 32 − 22 = 10; 10 − 1 = 9; 9 − 4 = 5; 5 + 1 = 6. These are not original review results.

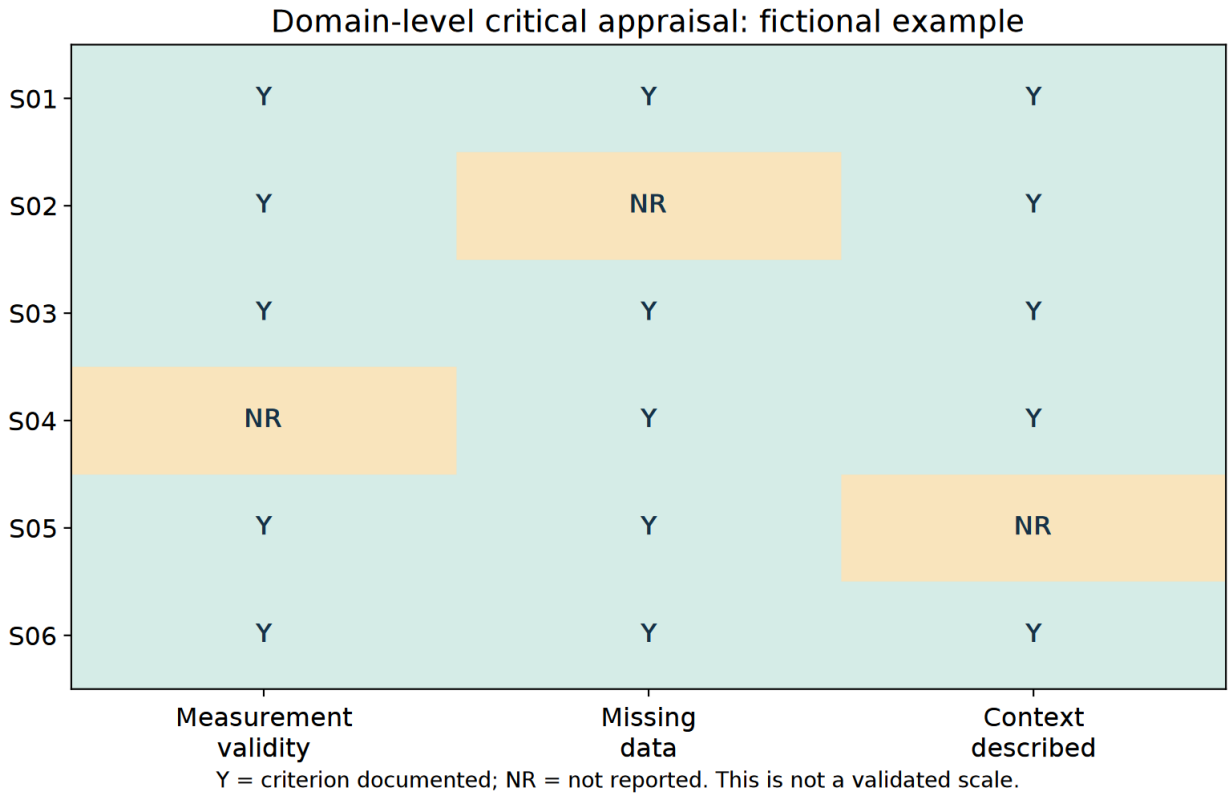


Fig. 6.3: Domain judgements in the fictional exercise. NR means not reported, not low risk. These teaching domains are not a validated scale and must not be summed. Original figure, CC BY 4.0.

trieval/eligibility and studies at the end [PMB+21]. Data are in the *case study*.

CHAPTER 7

PHASE 2C. DATA EXTRACTION AND SYNTHESIS

Extraction must be consistent: a form with the same fields for all studies, defined in the protocol and piloted on three or four papers before being used in earnest. Each field must serve to answer a specific research question; if it answers none, it is superfluous.

7.1 The extraction form

It has two blocks: generic data (unique identifier, full reference, year, venue and publication type, country or affiliation, study type, context, number of participants) and question-specific data (in this guide's case study on haptic simulators in surgery: metrics used, residents' level of experience, number of participants). Always add a free-text notes field, the critical-appraisal judgements and who extracted the data. The most practical option is a shared spreadsheet (Google Sheets, Excel) with one row per study and one column per field, or Parsifal's extraction form.

For categorical fields (contribution type, evaluation method, domain), define the classification scheme before extracting, with closed values. In systematic mapping studies, Petersen et al. (2008) propose building the scheme through *keywording*: reading the abstracts of the included studies, noting key concepts and grouping them into categories before classifying. Reusable schemes: the research type by Wieringa et al. (2006) (validation, evaluation, solution proposal, philosophical, opinion, experience) and contribution type (method, tool, model, metric, process).

Validate the extraction in the same way as the screening: a second person verifies data or extracts independently, using a justified sampling plan where necessary. Test-retest only checks within-person consistency. During full reading you may discover that a study is not relevant; it is excluded and the reason is recorded (Phelps & Campbell, 2012).

7.2 Traceability

You must be able to explain what happened to each paper in each phase and by which criterion it was excluded. Without that traceability, neither the paper's reviewer nor the thesis supervisor can trust the review. Most of that information does not fit in the paper, so it goes into supplementary material: a spreadsheet with all candidates and their decision at each phase, the search strings, the full extraction form and the quality checklist. Deposit it in Zenodo or OSF with a DOI and cite it from the paper; this is what PRISMA 2020 (item 27) and the Empirical Standards require.

7.3 Synthesis

The synthesis answers each question from the extracted data. It is advisable to follow four steps: present the data from the search and selection process (PRISMA diagram), tabulate the data of the included studies, represent the most characteristic data graphically and outline the answer to each question.

Method	When	How
Descriptive and tabular synthesis	Always; the basis of any mapping study	Tables of studies by category; counts and percentages; one table per question
Narrative synthesis	SLRs with heterogeneous studies	Group findings by theme or question and describe them with reference to the studies that support them; make contradictions explicit
Thematic synthesis or coding	Qualitative or mixed-methods studies	Code fragments, group codes into themes (Cruzes & Dybå, 2011, for software engineering)
Meta-analysis	Quantitative studies with the same variable and comparable designs	Effect sizes pooled with a random-effects model; heterogeneity (I^2); forest plots. Avoid *vote counting*
Mapping and visualisation	Systematic mapping studies	Trend charts by year, distribution by venue, *bubble plots* crossing two categories, Sankey diagrams from source to decision

7.4 Representing the data

The most useful figures (Hou et al., 2024, use several of them on 395 studies: distribution by venue and by year, annual evolution of model architecture types, counts of data strategies and *prompting* techniques, and a table of software engineering tasks by activity):

- **Trend chart**: number of studies per year, optionally by source; it shows whether the topic is growing or declining.
- **Table of authors and venues** with their number of publications where this answers an explicit question; venue indicators do not measure study quality; it answers "who" and "where".
- **Study × category cross-table** (an X per cell) for each question; it is the most compact way of showing which study contributes what.
- **Bubble plot** (Petersen et al., 2008; Kitchenham et al., 2015): two two-way tables sharing a nominal axis, with bubble size equal to the number of studies; it is the signature figure of a mapping study.
- **Sankey diagram** from sources to inclusion decision (SankeyMATIC) and **donut charts** by source and year to show the process.
- **PRISMA 2020 flow diagram** with the number of records at each step.

Use the same palette and the same category order in all figures, and avoid purposeless duplication; a data table can support verification while a figure supports interpretation.

7.5 Common mistakes

Common mistakes

- Extracting without a form and "sorting it out" afterwards.
- Columns that do not answer any question.
- Presenting the synthesis as a paper-by-paper list (the "shopping list" of the Empirical Standards) instead of by theme or question.
- Counting votes ("7 studies for, 3 against") as if it were a meta-analysis.
- Charts that repeat the table without adding insight.

7.6 From extracted data to a reasoned answer

1. Group by question and context, separating observations from authors' interpretations.
2. Code each passage with study, report and page; define categories in a codebook and document revisions.
3. Compare convergence, contradictions, designs and limitations. "Not reported" means neither "absent" nor zero.
4. Write a bounded conclusion citing the study IDs that support it and the conditions under which it might change.

In the *reproducible example*, S01 reports time and errors whereas S02 reports time only. The code "temporal efficiency" groups time; "execution safety" groups errors and force. This classification does not establish that time measures clinical competence or that one metric is superior. S02 omits handling of missing data and S04 omits measurement validity: both limitations must accompany interpretation.

For quantitative outcomes, do not count significant versus non-significant studies. Consider magnitude, direction, precision and compatibility. A random-effects model cannot by itself resolve incompatible conceptual or methodological differences [Cochrane].

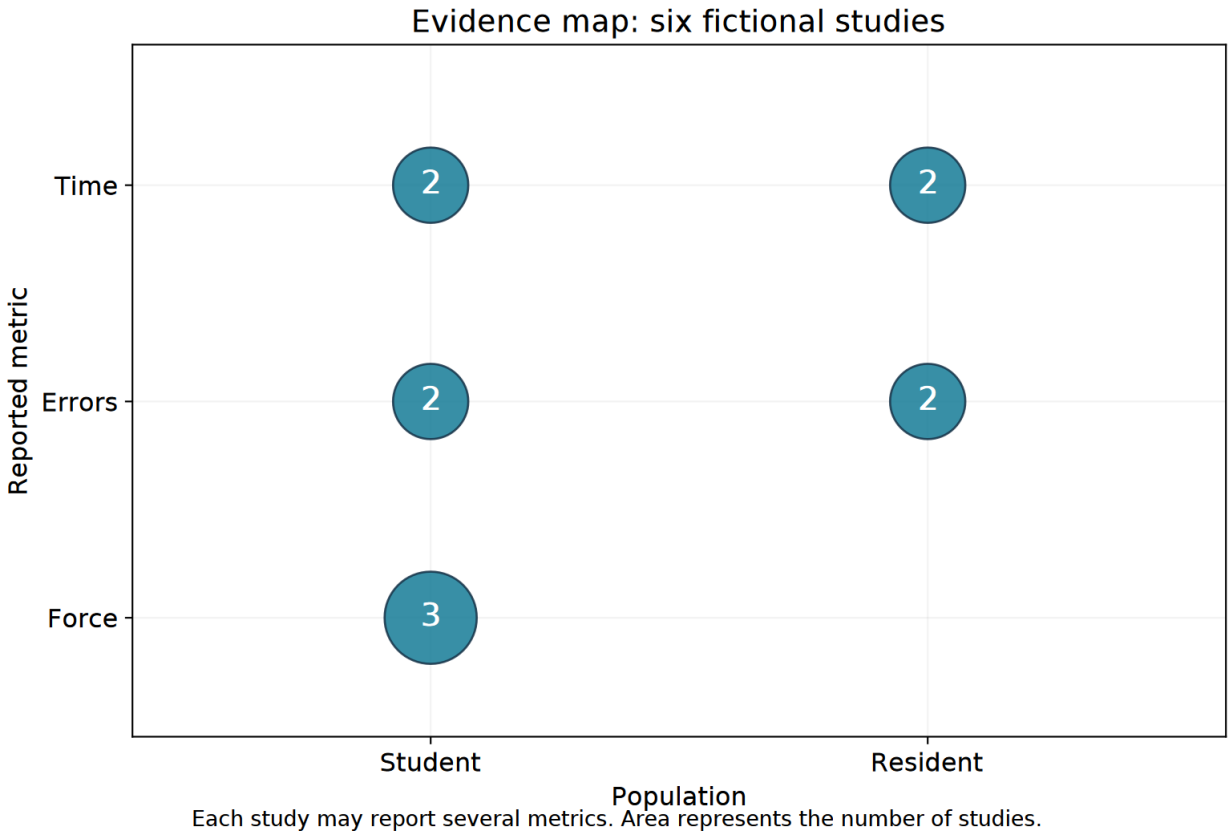


Fig. 7.1: Evidence map of the fictional exercise. Area and label indicate studies; one study can report several metrics. An empty cell alone does not establish research priority. Original figure, CC BY 4.0.

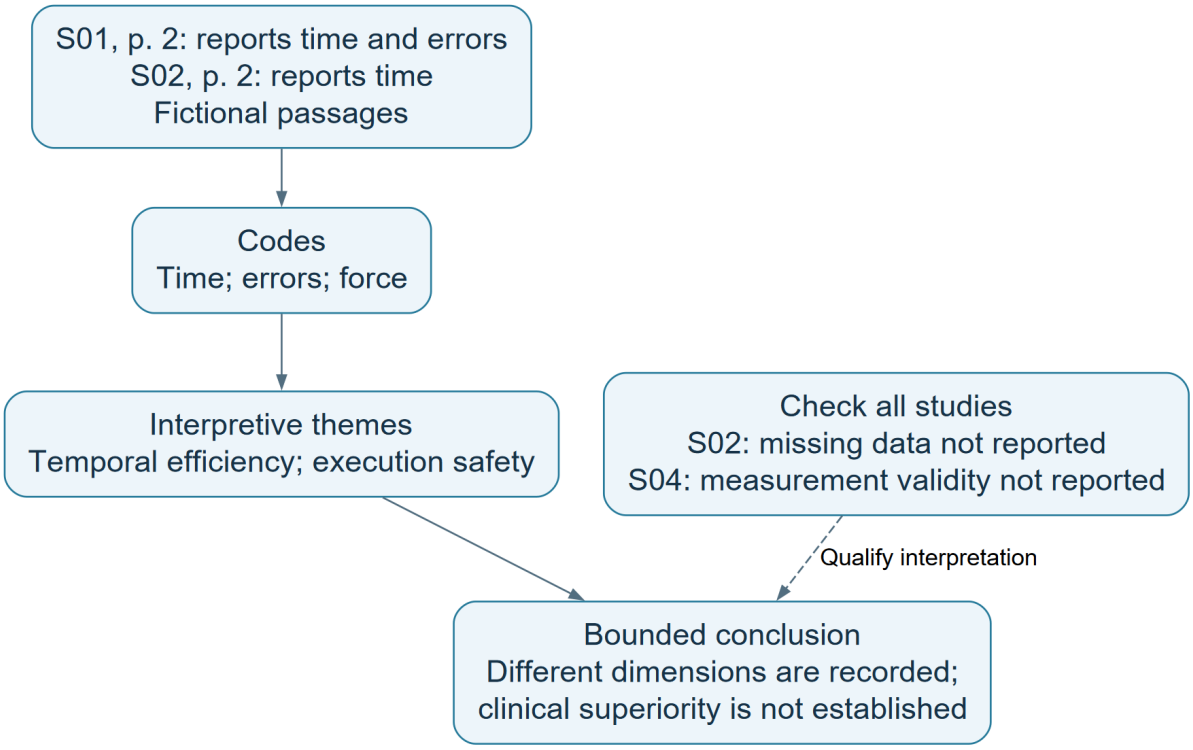


Fig. 7.2: From passage to theme and conclusion. Fictional passages: themes are interpretations requiring justification and checks against all studies. Original figure, CC BY 4.0.

CHAPTER 8

PHASE 3. WRITING AND PUBLICATION

An SLR paper has a fixed structure and reviewers look for it: the flow diagram, search strategies and limitations allow readers to assess transparency and rigour. Write with the checklist (PRISMA 2020 or SEGRESS) at hand and complete it before submitting; many journals request it as supplementary material.

8.1 Recommended structure (Kitchenham et al., 2015; PRISMA 2020)

#	Section	Content
1	Title	Topic and type of review (“systematic review”, “systematic mapping study”, “meta-analysis”)
2	Abstract	Structured: background, objectives, method (sources, dates, criteria, number of studies), results, conclusions (PRISMA 2020 for Abstracts, 12 items)
3	Introduction	Rationale (which reviews exist and why this one is needed), objectives and questions
4	Background	What is needed to understand the topic and the key concepts
5	Method	Where the protocol is; sources, strings and dates; eligibility criteria; selection process (how many reviewers, how disagreements were resolved, tools used); extraction; quality assessment; synthesis method. Decide what goes in the body and what goes in the appendix
6	Results	PRISMA diagram with the number of studies excluded at each phase; characteristics of the included studies; quality and biases; synthesis by question with tables and figures
7	Discussion	Answer to each question, implications for research and practice, and limitations
8	Conclusions	Overall interpretation and future work
9	Acknowledgements and declarations	Funding, conflicts of interest, use of AI if any
10	Appendices and supplementary material	Strings by source, extraction form, list of studies excluded at full text with reasons, quality checklist, extracted data

If you have carried out both a mapping study and an SLR, the paper usually has a results section for each; each section answers its own questions and uses charts, tables and visual explanations. Pautasso’s (2013) ten rules still hold: define the topic and audience, take notes while reading, keep the review focused but of broad interest, be critical and consistent, find a logical structure, make use of feedback, include your own research objectively, and stay up to date without forgetting older studies.

8.2 Limitations and threats to validity

An honest SLR declares its threats and what it did to mitigate them. The four classic ones in software engineering (Ampatzoglou et al., 2019) are:

- **Construct validity**: do the search terms and criteria capture the concept under study? Mitigation: control set, synonyms, piloting.
- **Internal validity**: were screening and extraction carried out without bias? Mitigation: two reviewers, kappa, registered protocol.
- **External validity**: do the results generalise? It depends on the coverage of sources, languages and years; the publication bias of the primary studies cannot be eliminated.
- **Reliability**: would another team obtain the same results? Mitigation: published strings, dates and data.

SEGRESS asks for this section to be called "limitations"; the content is the same.

8.3 Where to publish

Find recent reviews on your topic. Check publishers' scope, article types, review instructions, length and data and AI policies. Verify APCs, waivers and institutional agreements. Distinguish SJR and JCR, their categories and reference years.

Examples of topical fit: Information and Software Technology, Journal of Systems and Software and ACM Computing Surveys for computing; Computers & Education, Educational Research Review and EKS for education. These are not a ranking or acceptance guarantee.

Confirm whether the venue currently accepts your review type and retain source and checking date. When dividing a review across publications, justify distinct questions, disclose the shared corpus and avoid duplicating findings.

8.4 What to expect from peer review

The most common comments on an SLR: the search is not reproducible; an obvious source is missing; a single reviewer did the screening; there is no quality assessment; the synthesis is a list; threats to validity are not discussed; the paper is too long. Several of these can be anticipated with the checklist; the last one, by moving tables to the supplementary material. Bear in mind too that not all reviewers know the method well: justify the chosen critical appraisal and whether meta-analysis is appropriate.

Part III

Tools and AI

CHAPTER 9

TOOLS

Choose by task, traceability, institutional access, export and preservation. Test a small dataset before entering the whole review.

Need	Official options	What to check
References and duplicates	Zotero	Review merges; retain provenance
Screening and protocol	Rayyan, Parsifal, CADIMA, Covidence	Check roles, independence, disagreements and export
Active learning	ASReview	Validate ranking, stopping and auditing
Extraction	EPPI-Reviewer, JBI SUMARI	Or a spreadsheet: defined fields and source pages
Citation discovery	Semantic Scholar, ResearchRabbit, Connected Papers, Litmaps	Log steps, coverage and date
AI assistance	Elicit, Scite, NotebookLM	Verify documents and each extracted item
Bibliometrics	VOSviewer, bibliometrix	Does not replace critical synthesis
Visualisation	PRISMA2020	R, Python or a spreadsheet for other plots

9.1 Active learning and stopping rules

Published savings apply to particular datasets and configurations; they do not guarantee retrieval in your review [vdS+21]. Simulation on labelled data is not validation of a new review. Ranking records and leaving records unread are different decisions.

A run of 50–100 irrelevant records does not establish that no relevant studies remain. Justify stopping, auditing and residual risk; report unevaluated records and how they were handled. Volume alone does not determine the need for active learning.

9.2 Conditions to verify

Free plans, prices, versions and limits change. Consult official links and log date, version or plan, cost and applicable restrictions. Do not budget from historical figures. Check decision independence, role-based access, privacy and disagreement exports. Periodically preserve open-format copies of references, decisions, extraction and the data dictionary. The Systematic Review Toolbox helps locate alternatives by phase.

CHAPTER 10

GENERATIVE AI IN AN SLR: WHAT IT CAN DO AND HOW TO DECLARE IT

Generative AI is already part of the workflow of many reviews, but the evidence from 2023 to 2026 is clear: language models serve as **support** (second reviewer, pre-filter, extraction draft), not as a substitute for human judgement. The joint position statement of Cochrane, Campbell, JBI and CEE of November 2025 sums it up: authors remain responsible for everything, AI is used only if it does not compromise rigour, with human oversight, and any use that makes or suggests judgements is declared (Flemyng et al., 2025).

10.1 What the evidence says, phase by phase

Phase	What works	What does not, and evidence
Planning	Refining questions, suggesting synonyms and terms, reviewing the protocol against PRISMA-P	Nothing especially risky as long as you check what it proposes
Search	Generating search string variants and translating them between databases (Polyglot); preliminary search in Elicit or Undermind	Asking a chatbot for references: Chelli et al. (2024) measured fabricated reference rates of 28.6% for GPT-4 and 39.6% for GPT-3.5 (91.4% for Bard), with only 16 to 20% correct DOIs
Title and abstract screening	As automated screening assistance with a "sensitive" rule: Tran et al. (2024) obtain a sensitivity of 94.6 to 99.8%, missing 0 or 1 of 26 studies, at the cost of a specificity of 2 to 47%	As the sole reviewer: Guo et al. (2024), over 24,307 records, obtain a sensitivity for included studies of 0.76 (0.59 to 1.0 depending on the review). In software engineering, Felizardo et al. (ESEM 2024) obtain 75 and 86% accuracy with GPT-4 in two SLRs and conclude that outsourcing selection is not advisable; Thode et al. (IST 2025) reach a *recall* of 98 to 99% by combining two models, with a precision of 27%
Full-text screening	Promising with well-designed prompts (Khraisha et al., 2024)	Dependent on the prompt and the model version; unstable results
Data extraction	Draft of the extraction form from the PDF (Elicit, Covidence, an LLM with the PDF) that a human verifies field by field	"Moderate" reliability after adjusting for chance (Khraisha et al.); numbers and tables are frequently transcribed incorrectly
Synthesis and writing	Summarising your own notes, polishing the English, proposing a structure	Writing up findings without having read the studies; inventing citations

A pattern common to all studies: high sensitivity is only achieved by accepting many false positives, results vary with the prompt and the model version, and with older datasets there is a risk that the model has seen the reviews during training (Dennstädt et al., 2024). Model versions change every few months, so these figures go out of date; what does not go out of date is the obligation to validate on your own dataset.

10.2 Protocol for responsible use

1. Define tasks, model, version, prompts, parameters and oversight before use; document amendments.
2. Build a reference set assessed by two people with consensus resolution. Justify its size and composition: include enough relevant studies, uncertain cases and different document types. Separate prompt tuning from final validation.
3. Report true and false positives and negatives, sensitivity, specificity, precision and confidence intervals. **Neither 100–200 records nor a point sensitivity of 95% guarantees safety.** Recovering all five relevant studies still leaves substantial uncertainty about sensitivity.
4. Use the model to prioritise or propose decisions; a human verifies exclusions and resolves disagreements. An LLM does not count as a second independent human reviewer. Audit human–AI agreement too, since both may share

errors.

5. Retain inputs, outputs, versions, dates and final decisions; verify each extracted item against its source page. Revalidate after important changes to the model, prompt or corpus.
6. Respect licences and document processing conditions; use appropriate institutional services. Declare use in Methods, materials and the selection flow where applicable.

Stopping solely after 50–100 consecutive irrelevant records does not demonstrate complete evidence retrieval. The protocol must justify the risk of omissions and the checking procedure.

10.3 Guidelines and standards

- **RAISE** (*Responsible AI use in Evidence Synthesis*, Cochrane, 2025): three documents with recommendations by role, how to evaluate tools and how to select and use them; endorsed by more than 30 organisations. The Cochrane AI Methods Group (February 2025) coordinates its development.
- **CEE reporting guidance** (living document): for each AI system, name, version and developer; rationale and validation studies; automated tasks; own validation or the reason for not performing it; limitations; prompts and code as supplementary material; funding and conflicts of interest. It includes a template paragraph for the Methods section.
- **PRISMA 2020**: items 8 and 9 (selection and extraction) already ask you to state the automation tools used. The PRISMA-trAIce extension (JMIR AI, December 2025) proposes 14 items and a flow diagram that separates human and AI exclusions; it is a proposal without formal endorsement yet. Not to be confused with PRISMA-AI, which is for reviews of clinical AI studies.
- Cochrane's **Screen4Me** reduces records by 50 to 70% in reviews of trials, but only for Cochrane reviews.

10.4 What publishers require

Publisher	Policy
Elsevier (IST, JSS, C&E)	AI declaration in a dedicated section of the manuscript (tool, purpose, oversight); AI cannot be an author; basic grammar correction is exempt; AI used in the method is described in the Methods section
IEEE (TSE, TLT, Access)	Disclose in the acknowledgements the system and the affected sections (text, figures, code); IEEE Access additionally requires citing it in each section containing generated text
ACM (CSUR, TOCE)	AI cannot be an author; disclose generative use, usually in the acknowledgements; spelling/grammar checkers and translation are exempt
Springer Nature	Traffic light: language polishing (green), evaluative use or writing (amber: verify and declare), AI authorship or fabricated citations (red)

In all of them, reviewers are forbidden from uploading manuscripts to public tools. Also consult your university's research-integrity and AI policies; document the rules applicable when the work is performed.

These policy summaries are indicative and change: check the journal's current instructions at submission and log source and date. Validation percentages above refer to the historical models and datasets cited, not current performance of any LLM.

10.5 Example declaration paragraph

```
During title and abstract screening, GPT-4o (OpenAI, version 2026-05, via API) was␣
↪used as a second reviewer, with the prompt reproduced in supplementary material S3.␣
↪On a sample of 200 records screened independently by two authors, the model␣
↪achieved a sensitivity of 0.97 and a specificity of 0.61 (κ = 0.48). Disagreements␣
↪between the model and the first reviewer (n = 143) were resolved by the second␣
↪author. AI was not involved in extraction or synthesis. DeepL Write was used for␣
↪language polishing of the manuscript; the authors reviewed and are responsible for␣
↪all content.
```

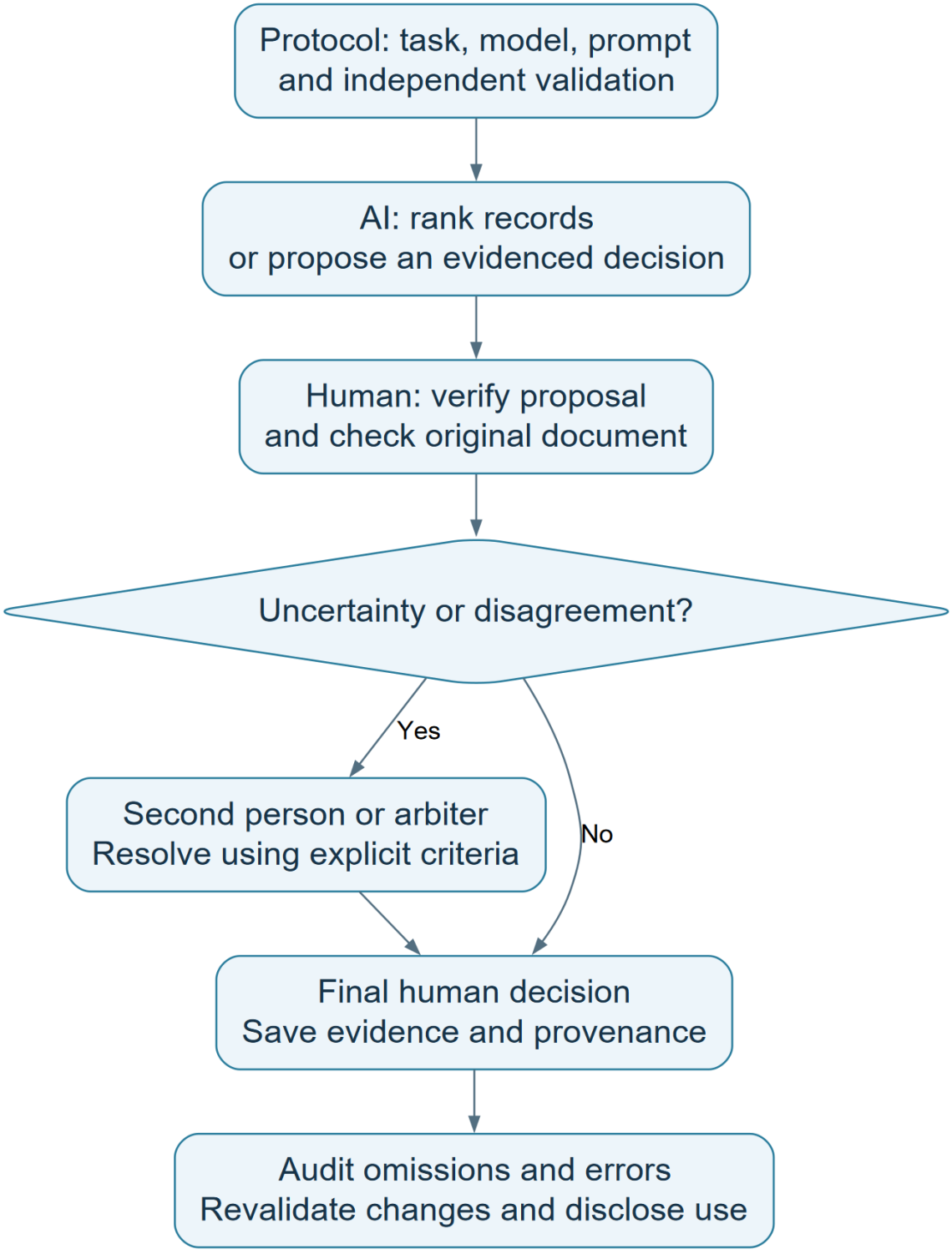


Fig. 10.1: Human–AI collaboration with human decisions and auditing. An automated proposal does not replace independent human review. Original figure, CC BY 4.0.

Part IV

Practice

CHAPTER 11

GUIDED CASE STUDY

The following case is a single-author SLR on metrics for assessing the performance of surgical residents using haptic simulators. It is small (fewer than 50 candidates) and therefore illustrates well the decisions and validations that need to be planned when there is no team.

11.1 1. Question and PICOC

Question: *In the field of surgery, what metrics exist to objectively assess a resident's performance using haptic simulators?*

Criterion	Value	Keywords
Population / Problem	Objective performance assessment; resident	"objective assessment"; performance; resident
Intervention	Haptic simulator	"haptic simulator"
Comparison	None	—
Outcome	Metrics	metric
Context	Surgery	surgery

11.2 2. Quasi-gold standard set

Before searching, identify by hand (or by asking an expert) a small set of papers that directly answer the question (Zhang, Babar & Tell, 2011, call it the *quasi-gold standard*). It will be used to extract keywords from their titles and abstracts and to validate the search string: a reasonable criterion is that the automated search should retrieve all known relevant papers indexed in each source (the historical example's 70% is not a general threshold) ; investigate each absence and revise the string where appropriate.

11.3 3. Sources and restrictions

Five or six sources combining general and specific ones: Web of Science, Scopus, ScienceDirect, IEEE Xplore and ACM DL (the last two specific to computing). Declared restrictions: search in title, abstract and keywords; English only; journal or conference papers only. The acknowledged limitation (only sources accessible from the University of León) is compensated for with complementary searching and grey literature.

11.4 4. Search string, iterated

```
Base string (WoS, 1 result):
("objective assessment" AND performance AND resident) AND "haptic simulator" AND␣
 ↪metric AND surgery

Extended string (WoS, 17 results):
("objective assessment" AND performance AND resident*) AND ("haptic simulat*" OR
 ↪"virtual reality") AND metric* AND surg*
```

The first version returned a single paper; truncating (*resident**, *metric**, *surg**) and adding a synonym for the intervention extended it to 17. Broadening brings in false positives; narrowing loses relevant papers (false negatives): the quasi-gold standard set is what tells you when to stop.

11.5 5. Selection criteria and phases

Inclusion: in English; journal or conference paper; defines assessment metrics; the metrics assess the performance of a resident or final-year student using a haptic or virtual simulator. Exclusion: another type of simulator (augmented reality, physical models); metrics on other aspects (realism, human factors, novice/expert discrimination).

With a single reviewer and few candidates, two phases are planned: (1) remove duplicates and review title and abstract, excluding if at least one exclusion criterion is met and moving on to the next phase in case of doubt; (2) download and review the full text with the same criteria. Validation by *test-retest* on a random sample; if the results differ, the criteria are revised.

11.6 6. Quality, extraction and synthesis

Dybå & Dingsøyr (2008) checklist scored Yes = 1, Partially = 0.5, No = 0, plus two specific questions (are the metrics clearly defined? is the population only students or residents?). Extraction in a spreadsheet with identifier, reference, metrics (type and description), resident's level of experience (none, final-year student, resident) and number of participants. Synthesis with a table of relevant papers, a trend chart by year and a bubble plot crossing level of experience and prior experience with simulators (X axes) against type of metric (Y axis).

11.7 How you would do it today, with what this guide adds

- Register the protocol on OSF before launching the search and cite its DOI.
- Document the search with the 16 PRISMA-S items and add backward and forward *snowballing* from the included studies.
- Plan independent human assessment, justify sampling and report disagreements before consensus.
- If an LLM proposes decisions, validate it on an independent set and disclose it as automation.
- Generate the diagram with the PRISMA 2020 app and complete the SEGRESS checklist before submitting.
- Deposit the screening sheet, extraction sheet and search strings on Zenodo.

11.8 The mistakes we see most often in PhD students

The mistakes we see most often in PhD students

1. Starting with the search string instead of the question.
2. Choosing a topic for which a recent SLR already exists and neither citing it nor justifying the new one.
3. Using only Google Scholar or only one database.
4. Not keeping the original exports or the search date, and being unable to reconstruct the PRISMA diagram.
5. Screening alone, without any validation, and not mentioning it as a limitation.
6. Confusing a mapping study with an SLR, so that the title promises what the method does not deliver.
7. Synthesising paper by paper instead of by question.
8. Submitting to a journal that no longer accepts reviews or whose format does not allow the necessary length.
9. Using AI to screen or extract without validating or declaring it.
10. Taking so long that the search becomes outdated; assess updates against the field's pace and submission date.

11.9 Complete reproducible exercise: fictional data

This exercise complements the historical case. **Records, studies S01–S06, participants, passages, pages and judgements are invented for teaching. They are not scientific findings and do not reconstruct the original review.**

Protocol: RQ1 asks which metrics are reported by population. Include residents or final-year students, a haptic/virtual simulator and at least one performance metric. EC1: wrong population; EC2: no metric. Fictional reviewers A/B use the same criteria; their agreement does not estimate expected agreement in an actual review. The three appraisal domains are illustrative, with no total score. No AI is used.

Selection: Scopus contributes 24 records and IEEE Xplore 16. Eight duplicates are removed, 32 records screened and 22 excluded. Ten reports are sought: one unavailable and nine assessed. Two are excluded for EC1 and two for EC2. Five database reports and one citation-search report are included: six reports and six studies. The citation branch contains one report sought, retrieved and eligible. R28 is unavailable: log access/contact attempts in a real review; it is not a content-based exclusion.

Study	Population	Participants	Metrics (fictional page 2)
S01	Resident	24	Time; errors
S02	Resident	18	Time
S03	Student	30	Errors; force
S04	Student	16	Time; force
S05	Resident	22	Errors
S06	Student	20	Time; errors; force

Answer to RQ1: time and errors each appear in four studies; force in three. All three student studies report force; two report time and two errors. Among residents, two report time and two errors. These are reporting frequencies, not effectiveness comparisons. Fast performance can include errors: no metric is shown to be superior and participants are not pooled as one sample.

Appraisal and interpretation: S02 does not report missing-data handling; S04 does not report measurement validity; S05 insufficiently describes context. Other cells document the criterion. "Not reported" requires caution rather than

proving faulty methods. Retain these limitations when grouping time as temporal efficiency and errors/force as execution safety.

Downloads: `Full screening log` · `Extraction` · `Codebook` · `Counts` · `Full package`.

Reproduction: install `requirements-figures.txt` in `.venv` and run `scripts/generate_teaching_assets.py` from the repository root. It checks counts and plots downloadable extraction data. Then run `scripts/render_diagrams.py` for Graphviz sources. Column names and IDs remain in English for stability; values and figures are translated.

Check your understanding: why do 11 metric occurrences not mean 11 studies? Why is R28 not added to the four exclusions? Categories overlap and R28 could never be assessed.

CHAPTER 12

TEMPLATES AND CHECKLISTS

Copy these templates into your own document or spreadsheet and fill them in order. They are aligned with PRISMA-P, PRISMA-S and SEGRESS, so that when you finish you will have almost everything the reporting checklist will ask for.

12.1 Protocol template

```
# Systematic review protocol: <title>
Version: 1.0 · Date: <yyyy-mm-dd> · Registration: <OSF/Zenodo DOI>
Authors and roles: <reviewer 1, reviewer 2, arbiter>

## 1. Rationale
- Previous reviews located (search string and date of that search) and why they are␣
 ↪not sufficient
- Whom this review serves and how

## 2. Objectives and questions
- Review type: SLR / mapping / scoping / MLR
- PICOC
- MQ1… / RQ1… (each with the extraction field that answers it)

## 3. Search strategy
- Sources (name, platform, reason)
- Generic search string and adaptation per source
- Restrictions: fields, years, languages, document types (justified)
- Control set (list of papers) and retrieval threshold
- Snowballing: backward / forward, stopping criterion
- Grey literature: sources and stopping criterion

## 4. Eligibility criteria
- Inclusion (IC1…, joined by AND)
- Exclusion (EC1…, joined by OR)

## 5. Selection process
- Phases (duplicates, title/abstract, full text)
- Reviewers per phase, calibration, disagreement resolution, agreement measure
- Tools (reference manager, screening, active learning, AI) and how they are validated

## 6. Quality assessment / risk of bias
- Tool by design, domains, rationales and use in synthesis; justified exclusions and␣
 ↪sensitivity analyses

## 7. Data extraction
```

(continues on next page)

(continued from previous page)

```
- Form (fields, allowed values), piloting, validation

## 8. Synthesis
- Method per question (tabular, narrative, thematic, meta-analysis), planned figures

## 9. Reporting and dissemination
- Checklist (PRISMA 2020 / SEGRESS), target venue, supplementary material

## 10. Schedule and resources

## 11. Protocol change log
| Date | Change | Reason |
```

12.2 Extraction form (spreadsheet columns)

Block	Fields
Identification	ID (S01…), full citation, DOI, year, venue, type (journal, conference, chapter), country of first affiliation, source it comes from
Design	Research type (Wieringa: validation, evaluation, solution proposal, philosophical, opinion, experience), empirical method (experiment, case study, survey, quasi-experiment), context (academic, industrial), participants and number
Content	One field per research question, with closed values where possible; contribution type (method, tool, model, metric, process)
Quality	Domain judgement, rationale, page and effect on synthesis
Management	Extracting reviewer, date, notes, relevant verbatim quotes with page

12.3 Screening sheet (one row per record)

Record ID, report ID, study ID, source, title, year, report retrieved (yes/no), duplicate (yes/no and of which), phase 1 decision reviewer A, phase 1 decision reviewer B, final phase 1 decision, phase 2 decision, exclusion criterion applied, origin (search / snowballing / grey). Count records, reports and studies separately; link multiple reports of one study. Record retrieval attempts and reasons for unavailable reports too.

12.4 Abridged reporting checklist (PRISMA 2020 + SEGRESS)

- The title states that it is a systematic review or a mapping study
- Structured abstract with sources, dates, criteria, number of studies and conclusion
- Rationale with previous reviews cited
- Numbered questions and PICOC
- Protocol registered and linked; deviations explained

- All sources with full search string, filters and date (PRISMA-S)
- Explicit inclusion and exclusion criteria
- Selection process: phases, reviewers, agreement, automation or AI tools
- PRISMA 2020 flow diagram with numbers at each step
- List of studies excluded at full text with reason (supplement)
- Quality or risk-of-bias instrument and results per study
- Extraction form and extracted data available
- Synthesis by question, with tables and figures that do not duplicate each other
- Limitations or threats to validity with mitigations
- Implications for research and practice
- Funding, conflicts of interest and AI declaration
- Materials in a repository with a DOI

12.5 Indicative schedule (PhD SLR, 300 to 1,500 candidates)

Phase	Typical duration	Deliverable
Preliminary search and check for existing reviews	2 weeks	List of previous reviews; decision to proceed
Protocol and review by the supervisor	3 to 4 weeks	Protocol v1.0 registered
Search, export and deduplication	1 to 2 weeks	Zotero library; search table
Title and abstract screening (with calibration)	2 to 4 weeks	Screening sheet; kappa
Full text and snowballing	4 to 6 weeks	List of included and excluded studies with reason
Quality and extraction	4 to 8 weeks	Validated extraction sheet
Synthesis and figures	3 to 4 weeks	Tables and figures per question
Writing, checklist and submission	4 to 6 weeks	Manuscript and supplementary material

Total: 6 to 9 months part-time. Assess updates against the field's pace and elapsed time; twelve months is an indicative warning, not a universal threshold.

12.6 Downloadable templates

- `Editable protocol`
- `Blank screening log`
- `Blank extraction form`
- `Search log`
- `Amendment log`
- `Editable checklist`
- `All templates and worked example`

Part V

References

CHAPTER 13

REFERENCES AND RESOURCES

13.1 Methodological guidelines

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- Kitchenham *et al.* [KBB15] — Publisher
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- Petersen *et al.* [PFMM08] — PDF
- Petersen *et al.* [PVK15] — EBSE
- Wohlin [Woh14]
- Garousi *et al.* [GFM19] — arXiv
- Ralph and others [R+21a] — Systematic Reviews standard
- Zhang *et al.* [ZBT11]
- Dybå and Dingsøyr [DD08] — PDF
- Grant and Booth [GB09]
- Mengist *et al.* [MSL20]
- Codina [Cod15] — Website
- Pautasso [Pau13]
- Petticrew and Roberts [PR05]
- Higgins, J. P. T., et al. (eds.). *Cochrane Handbook for Systematic Reviews of Interventions*. Current version

13.2 Reporting and registration

- Page *et al.* [PMB+21] — BMJ · Spanish version · Checklist · Flow diagram
- Rethlefsen and others [R+21b] — Article
- Tricco and others [T+18] — Website
- Moher and others [M+15] — Website
- Haddaway and others [H+22] — App
- Registries: PROSPERO · OSF Registries · INPLASY · Zenodo

- Huotala *et al.* [HKM25] — arXiv

13.3 AI in reviews

- Flemyng and others [F+25] — Text
- RAISE: Responsible AI use in Evidence Synthesis [RAISERAuiESynthesis25] — OSF · Cochrane
- CEE. Artificial intelligence reporting guidance. Website
- Holst and others [H+25] — Article
- Guo and others [G+24] — Article
- Tran and others [T+24] — Abstract
- Khraisha and others [K+24] — arXiv
- Dennstädt and others [D+24] — PDF
- Felizardo and others [F+24] — Programme
- Huotala and others [H+24] — arXiv
- Thode *et al.* [TIM25] — Abstract
- Chelli and others [C+24] — Article
- van de Schoot and others [vdS+21] — Article
- Gusenbauer and Haddaway [GH20]
- Li [Li21] — arXiv

13.4 Examples and readings in Spanish

- Conde González, M. Á., & Rodríguez Sedano, F. J. *Elaboración de una revisión sistemática de la literatura*. Universidad de León and GRIAL, USAL.
- García-Peñalvo [GP17] — Repository
- Hou *et al.* [HZL+24] — arXiv · Materials
- Ferreras-Fernández *et al.* [FFMRGPMV16]
- Humanante-Ramos *et al.* [HRGPCG17]
- Briz Ponce [BP16] — Handle
- Gisbert and Bonfill [GB04]
- Martín Rodero [MR14] — Handle
- Sáenz [Sae01]
- Boote and Beile [BB05]

13.5 Other resources

- Systematic Review Toolbox: catalogue of tools by phase.
- EQUATOR Network: all PRISMA extensions and guidelines under development.
- TERA tools (formerly SR-Accelerator): Polyglot, Deduplicator, Word Frequency Analyser.
- Cochrane Training: free methodology courses.

13.6 Critical appraisal and certainty

- Hong *et al.* [HPF+18]
- Shea *et al.* [SRW+17]
- Schunemann *et al.* [SBA+23]

Part VI

Information

CHAPTER 14

ABOUT

This book is a practical guide to conducting a systematic literature review (SLR) or a systematic mapping study in engineering and computer science, aimed at PhD students and researchers undertaking their first review. It was written as support material during the PhD and covers the current standards, an overview of tools and a section on the responsible use of generative AI.

Citation information is gathered on the *How to cite* page.

14.1 How this book is made

This site is written in Markdown files and converted to HTML and PDF using tools from the TeachBooks, Jupyter Book and MyST Markdown ecosystem, based on the TeachBook USAL template by Álvaro Lozano Murciego and André Filipe Sales Mendes.

The source files are in a public GitHub repository:

- https://github.com/danihiglesias/guia-slr

14.2 About the author

Daniel H. de la Iglesia University of Salamanca

- Email: danihiglesias@usal.es

CHAPTER 15

CREDITS AND LICENSES

15.1 Project credits

This book has been built from the TeachBook USAL template (Lozano Murciego and Sales Mendes, 2026, CC BY 4.0) with TeachBooks, on top of the Jupyter Book and MyST Markdown ecosystem.

TeachBooks provides documentation, components and workflow patterns that make it easier to create open, interactive and reproducible teaching books.

15.2 Acknowledgements

We thank the TeachBooks project and Delft University of Technology (TU Delft) for their work in developing open tools for educational books. The TeachBooks manual credits page serves as a reference for explicitly acknowledging the tools, communities and institutional support on which this material relies.

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15.5 Code and tools

The project makes use of external tools and libraries, including:

- TeachBooks
- Jupyter Book
- MyST Markdown

Each tool retains its own license.

15.6 Figures and data in this edition

Original teaching figures and fictional data: CC BY 4.0. Sources: `diagram_sources/es/`, `diagram_sources/en/` and `scripts/generate_teaching_assets.py`. The PRISMA flow adapts Page et al. (2021), with invented counts and reorganised labels; the original templates are CC BY 4.0. Captions identify authorship or adaptation. The matrix illustrates judgements rather than reproducing a validated tool.

CHAPTER 16

HOW TO CITE

If you use this material, please cite it as follows.

16.1 Recommended citation (APA 7th edition)

de la Iglesia, D. H. (2026). *Guide to conducting a systematic literature review (SLR)*. University of Salamanca. https://github.com/danihiglesias/guia-slr

16.2 BibTeX

You can also use the downloadable BibTeX file: `citation_en.bib`.

```
@book{de_la_iglesia_guide_slr_2026,
  author = {de la Iglesia, Daniel H.},
  title = {Guide to conducting a systematic literature review (SLR)},
  year = {2026},
  publisher = {Universidad de Salamanca},
  url = {https://github.com/danihiglesias/guia-slr}
}
```

Note

When the book is deposited in Zenodo, this page will be updated with the definitive DOI.

CHAPTER 17

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This is the unified book bibliography, generated from one BibTeX database. The resources chapter groups readings by topic and links their sources.

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